

LinearSolver

September 15, 2026

This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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1 Linear solvers

We demonstrate solving a dense system $Ax=b$ by using different linear solvers.

```
[1]: import casadi as ca
     from numpy import *
     import time
```

```
[2]: n=100
```

We generate $A \in \mathbf{R}^{n \times n}$, $x \in \mathbf{R}^n$ with $n = 100$

```
[3]: A=ca.DM([[cos(i*j)-sin(i) for i in range(n)] for j in range(n)])
     x=ca.DM([tan(i) for i in range(n)])
```

We generate the b vector:

```
[4]: b= A @ x
```

Commented out pending completion #1615

2 We demonstrate the LinearSolver API with CSparse:

```
s = LinearSolver("s", "csparse", A.sparsity())
```

3 Give it the matrix A

```
s.setInput(A,"A") # Do the LU factorization s.prepare()
```

4 Give it the matrix b

```
s.setInput(b,"B")
```

5 And we are off to find x...

```
s.solve()
```

```
x__ = s.getOutput("X")
```

6 By looking at the residuals between the x we knew in advance and the computed x, we see that the CSparse solver works

```
print "Sum of residuals = %.2e" % norm_1(x-x__)
```

7 Comparison of different linear solvers

8 =====

```
for solver in ("lapacklu","lapackqr","csparse"): s = LinearSolver("s", solver, A.sparsity()) # We create a solver
```

```
s.setInput(A,"A") # Give it the matrix A
```

```
t0 = time.time() for i in range(100): s.prepare() # Do the LU factorization pt = (time.time()-t0)/100
```

```
s.setInput(b,"B") # Give it the matrix b
```

```
t0 = time.time() for i in range(100): s.solve() st = (time.time()-t0)/100
```

```
x__ = s.getOutput("X")
```

```
print "" print solver print "=" * 10 print "Sum of residuals = %.2e" % norm_1(x-x__) print "Preparation time = %0.2f ms" % (pt*1000) print "Solve time = %0.2f ms" % (st*1000) assert(norm_1(x-x__)<1e-9)
```

9 Note that these